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Research Article submission

Dear Editors of IEEE Access,

Please consider our manuscript, “Protocol-Constrained Residual Estimation for Rigid-Body Inference: Strong Baselines and Visual Controls,” by Hyojin Park, Nam-Hyun Yoo, and Jinhong Yang, for publication as a **Research Article**.

The study examines whether a compact visual residual can add predictive information after a strong numerical estimator has already used a short motion observation. The method combines fixed robust estimation, numerical residual adaptation, and a bounded frozen-feature residual with protocol-specific permissions. Its contribution is the tested allocation of roles and the measured increment, rather than a claim that hybrid estimation or VLM-based physical inference is new in isolation.

The manuscript presents a protocol-constrained residual estimator evaluated against nonlinear classical anchors and strengthened numerical/visual controls. On 72 geometry families, the locked full estimator reduces held-out-action trajectory error by 20.30% relative to profiled-dispersion MAP (1.82315 to 1.45312 mm). A nuisance-marginal Laplace baseline and a VLM-free population-prior estimator make the baseline scope explicit.

The added 87-dimensional numerical-only cell and dimension-matched Lucas–Kanade, raw-pixel and V-JEPA controls separate residual-estimator performance from claims about a particular visual representation. The Qwen-derived view does not show a resolved advantage over classical tracking. Forty newly generated geometry families support a separate calibration analysis, and the real-video score is rerun after correcting its feature-unit contract. All comparisons reusing evaluated cohorts are labeled exploratory, with multiplicity-adjusted results supplied.

A public pushing evaluation retains 4.38% material-pair improvement with both friction-condition axes held out within the same object and scene assets. The paper preserves the original failed development criterion and unsuccessful real transfer, without making those boundaries its central contribution. We believe the concrete estimation design, strong controls and reproducible evidence are relevant to IEEE Access. The initial-submission package contains source, supplementary analyses, figures and versioned verification materials at GitHub release v0.4.0; funding and AI assistance are disclosed in the manuscript.

Thank you for considering this work.

Sincerely,
Jinhong Yang
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